

Divyansh Bansal

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New Delhi, India | Open to relocating to the United Kingdom

Professional Summary

AI Engineer with 1 year of production experience building **LLM-powered agentic systems** on **Azure OpenAI**. Deployed a **five-agent reconciliation platform** to production within Amdocs' **AT&T account** at enterprise scale, automating **40+ hours/week** of manual operations. Built retrieval-grounded **NL-to-SQL pipelines**, hermetic **CI-gated evaluation harnesses**, and self-improving **agent memory systems**. Published researcher in Wiley's *Software: Practice and Experience*. Databricks-certified in Data Engineering and Generative AI.

Experience

AI Software Engineer | Amdocs – AT&T

Sept 2025 – Present

New Delhi

- Architected a **five-agent reconciliation platform** (custom **Python/FastAPI** runtime, **Azure OpenAI GPT-4.1**); all five agents **deployed and live in production**, automating **40+ hours/week** of manual work.
- Designed per-agent **cost/latency budgets**, **deterministic fallbacks**, and **human-in-the-loop checkpointing**; presented architecture to **40+ cross-functional stakeholders** to secure production sign-off.
- Built a unified async **LLM gateway**: **structured-output generation**, bounded **self-repair**, **circuit-breaker failover**, and per-session **cost-budget gating**.
- Designed a retrieval-grounded **NL-to-SQL pipeline** with **schema pruning** and dry-run **self-correction**; cut prompt tokens **~59%** and latency **~70–85%** vs. baseline.
- Built a hermetic, **CI-blocking evaluation gate** (8 offline suites, zero external dependency), validated at **1,000-reconciliation scale** across 6 failure archetypes.
- Implemented **four-plane agent memory** (Redis, PostgreSQL, embeddings, procedural library) driving a self-improving correction loop.
- Leading migration of a legacy Oracle PL/SQL subscriber-event pipeline to **Azure Databricks**: parsing inbound events, applying business rules, and populating gold-layer tables for downstream consumption; rebuilding the logic in **PySpark** on a bronze/silver/gold **medallion architecture** with **Unity Catalog**.
- Optimised stored procedures processing **75M records/day**, cutting latency from 45 min to 12–15 min (**~73%**).
- Built an **MCP-based automation agent** integrated with **Apache Airflow**, enabling on-demand pause, resume, and stop control over pipeline jobs during infrastructure maintenance windows.
- Mentored **15+ engineers** through hands-on **Agentic AI workshops** on **LangChain** and **LangGraph** production patterns.

Co-Founder & Product Architect | DTU ERP Portal

Oct 2024 – July 2025

New Delhi

- Architected and deployed an **ML-powered ERP platform** serving **1,200+ users** across the university.
- Reduced accreditation-preparation time from 25 to 7 days (**~72% reduction**) through intelligent workflows and **semantic search**.
- Built a **semantic-retrieval pipeline** using **vector embeddings** and **ChromaDB** for document discovery.
- Developed **8+ modules** for accreditation automation, analytics dashboards, and a student-alumni network.
- Led architecture decisions, frontend development, and **cross-stakeholder adoption**; platform now **funded by DTU** for continued development.

Education

Delhi Technological University (DTU), New Delhi

2021 – 2025

B.Tech, Electronics and Communication Engineering | **CGPA 8.06 / 10.00**

Technical Skills

AI & LLM Engineering: Azure OpenAI (GPT-4.1), LangChain, LangGraph, MCP, RAG, Agent Orchestration

Retrieval & Search: FAISS, ChromaDB, Semantic Search, Vector Embeddings, BM25, Cross-Encoder Reranking

Machine Learning & Deep Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, LoRA/PEFT Fine-tuning, SHAP Explainability, Statistical Hypothesis Testing

Natural Language Processing: Hugging Face Transformers, NL-to-SQL, Text Generation, Structured Output Generation, Sequence Modelling, Tokenisation

Data & Infrastructure: Python, FastAPI, Docker, GCP Cloud Run, PostgreSQL, Redis, Apache Airflow

MLOps & Evaluation: MLflow, DagsHub, CI/CD Eval Gates, Drift Monitoring (PSI/KS), Model Versioning

Tools & Languages: Git, GitHub Actions, Claude Code, GitHub Copilot, Pandas, NumPy, Streamlit

Projects

IntelliCode: Hybrid RAG & AST Code Intelligence | HF ZeroGPU [Demo](#) | [Code](#)

- Built a **hybrid Retrieval-Augmented Generation pipeline** combining dense (**FAISS**) and sparse (**BM25**) retrieval with **reciprocal rank fusion** and **cross-encoder reranking**.
- Deployed **Qwen2.5-3B** for generation on **Hugging Face Spaces** (ZeroGPU).
- Built a labelled **evaluation harness** (MRR@5, Recall@k, NDCG) quantifying each retrieval stage's lift; **121 tests** regression-gated in **GitHub Actions CI**.
- Developed an async-aware **AST analyser** spanning **12 anti-pattern classes** and a **security scanner** (injection, weak crypto, hardcoded secrets).

Loan Risk Intelligence: Credit Default Prediction | ~1.8M LendingClub loans [API](#) | [Dashboard](#) | [Code](#)

- Engineered a two-stage stacking ensemble (**XGBoost** + **PyTorch** feed-forward) achieving Test **AUC 0.9184**; validated via **McNemar's test** ($\chi^2=194.8$, $p<0.0001$), audited with **SHAP**.
- Diagnosed a 2016 **underwriting regime shift** (default rate 9.4% → 2.4%) and corrected temporal leakage with **year-stratified splits**.
- Deployed on **GCP Cloud Run** (FastAPI, Docker) with live per-request **TreeSHAP** explanations, **Streamlit** dashboard, and **PSI/KS drift monitoring**.

Decoder-Only Transformer (From Scratch) | ~10.7M parameters

- Built a **decoder-only transformer** from scratch (~**10.7M parameters**) with no framework abstractions: implemented multi-head self-attention, positional encoding, layer normalisation, and autoregressive generation in raw **PyTorch**.
- Validated training convergence and generation quality on text corpora, demonstrating end-to-end understanding of transformer internals.

Publication

“Design and Implementation of a Multiuser Enterprise Resource Planning Solution for Higher Education Institutions: Enhancing Accreditation and Administration.”

Software: Practice and Experience (Wiley) | DOI: [10.1002/spe.70060](https://doi.org/10.1002/spe.70060)

Certifications

Databricks Data Engineer Associate | **Databricks Generative AI Engineer Associate**